

AI AGENT INFRASTRUCTURE

Agents Tools

Memory, learning, and intelligent skill curation for autonomous AI
agents

Quoth v3.4.0 • Skill Registry v1.0.0

THE PROBLEM

Agents Start From Zero, Every Time

No Memory

Every session erases all learned context. Agents repeat the same mistakes across projects.

No Feedback Loop

Successes and failures vanish at session end. No way to learn from outcomes.

Manual Skill Selection

Skills installed manually with no relevance matching. No optimization per project.



CONTEXT LOST EACH SESSION

MEMORY & LEARNING

Quoth

Self-learning platform that captures agent trajectories, distills patterns, and injects high-confidence knowledge.

- 9 hook bindings, 22 MCP tools
- 3-stage LLM pipeline (JUDGE → DISTILL → CONSOLIDATE)
- Bayesian scoring with Thompson sampling
- Unified injection: patterns + docs ranked together
- Graded reward signal (0.0 – 1.0)

KNOWLEDGE CURATION

Skill Registry

Centralized skill management that matches, forks, and AI-rewrites skills per project context.

- Semantic search via pgvector (1536d)
- Per-project fork & AI rewrite pipeline
- MCP server + REST API (Hono + Bun)
- Quoth patterns feed optimization queries
- Drift detection for stale forks

Architecture & Data Flow



9 Bindings, Zero API Calls

HOOK EVENT	COMMAND	ACTION
UserPromptSubmit	route	Route task to optimal agent, inject patterns ≥ 0.1
SessionStart	session-restore	Init intelligence graph, inject max 3 patterns ≥ 0.6
SessionEnd	session-end	Consolidate graph, recompute PageRank, Bayesian update
PreCompact	session-end	Same as SessionEnd (pre-context-compression)
PostToolUse	trajectory-capture	Capture Bash/Write/Edit/Agent to JSONL
PostToolUse	post-edit	Record Write/Edit for intelligence
PreToolUse	pre-bash	Block rm -rf /, fork bombs
SubagentStart	subagent-start	Inject domain patterns via additionalContext
SubagentStop	post-task	Bayesian +1 alpha on matched patterns

Centralized, Versioned, Intelligent

Semantic Search

pgvector HNSW with 1536d embeddings via Vercel AI Gateway. Find skills by meaning.

Per-Project Forks

Fork any skill, scope to project. AI rewrites content using tech stack + CLAUDE.md context.

Jinja2 Templates

Skills as markdown or Jinja2 templates with dynamic variables rendered at runtime.

Drift Detection

Detect when base skills change post-fork. Re-optimize without losing customization.

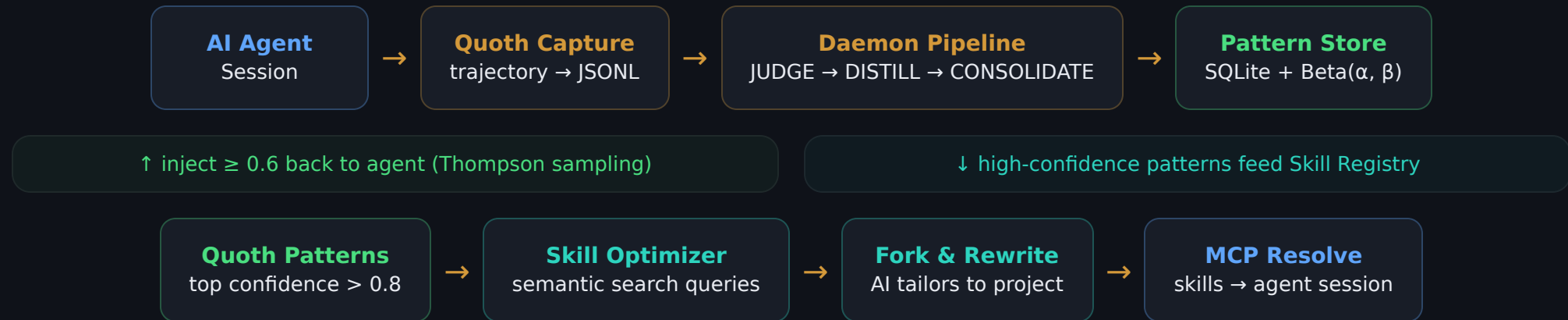
Agent Binding

Skills assigned to agents (OpenClaw, Claude Code, Quoth, Agentic). Resolved via MCP.

Optimization Pipeline

Scan project → profile tech stack → semantic match → fork top skills → AI rewrite.

How They Reinforce Each Other



Tech Stack Comparison

LAYER	QUOTH	SKILL REGISTRY
Runtime	Node.js (Claude Code plugin)	Bun + Hono
Database	SQLite (WAL) + HNSW	Neon Postgres + pgvector
Embeddings	MiniLM-L6-v2 (384d, local, \$0)	text-embedding-3-small (1536d)
LLM Pipeline	Gemini 2.5 Flash + Lite + Haiku 4.5	GPT-4o-mini via AI Gateway
API	MCP stdio (22 tools)	REST (Hono) + MCP stdio (6 tools)
Cloud	Next.js 16 on Vercel (Neon + pgvector)	Bun on Vercel (Neon)
Auth	Clerk + qth_* API keys (SHA-256)	X-API-Key header
Async Jobs	fs.watch + timers + SIGUSR2	QStash (Upstash)

AT A GLANCE

Ecosystem Metrics

22

QUOTH MCP TOOLS

6

SKILL REG TOOLS

9

HOOK BINDINGS

246

TESTS PASSING

100+

SKILLS INDEXED

4

AGENT TYPES

Self-Healing

- Exception handling — daemon never crashes
- Stale lock auto-cleanup from previous crashes
- HNSW rebuild from DB on corruption
- Exposure-based decay (30d archive, 600 cap)
- Stale session detector (10min, synthetic summaries)
- Auto doc maintenance (nightly hash scan + LLM update)

Graceful Degradation

- ~~Gemini API down~~ → Outcome-based judging
- ~~Embedding model fail~~ → Keyword matching
- ~~Cloud unavailable~~ → Local learning continues
- ~~Redis unavailable~~ → Rate limiting disabled
- ~~Reranker down~~ → Vector similarity order

Completed & Planned

SYSTEM	FEATURE	STATUS
Quoth	22 MCP tools, 9 hooks, local + managed daemon, CLI wizard	DONE
Quoth	Unified injection — patterns + docs in single ranked pipeline	DONE
Quoth	V2 hierarchical Thompson sampling + graded reward + SNIPS	DONE
Quoth	Local MiniLM-L6 384d (\$0) + batch JUDGE (Gemini) + doc re-indexing	DONE
Skill Reg	REST API + MCP server, semantic search, skill CRUD	DONE
Skill Reg	Project optimization (scan → profile → search → fork → rewrite)	DONE
Skill Reg	Drift detection, skill groups, Jinja2 rendering	DONE
Ecosystem	Quoth patterns as Skill Registry search signal	LIVE
Ecosystem	Usage dashboard + tier system (free/pro/team)	PLANNED
Ecosystem	Cross-org pattern sharing + skill marketplace	PLANNED

GET STARTED

Build Agents That Learn & Know

Quoth – quoth.triquail.dev • Skill Registry – Hono + MCP + Neon